

Bruno Marcuche

Platform Architect

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Summary

Platform Architect and technical leader who builds the systems other teams run on. I've scaled infrastructure across on-prem, hybrid, and cloud, and automated deployment and operations for thousands of Linux and Windows instances. Most recently I architected an internal AI agent platform, a custom semantic router orchestrating specialized LLM agents, that has routed 40,000+ ops requests, 83% of them without LLM reasoning. I lead ops and SRE teams, drive observability with OpenTelemetry and PagerDuty, and turn slow, manual operations into fast, repeatable automation.

Professional Experience

08/2026 to present
Boulder, CO

Platform Architect AssetWorks

- Architect and operate the hosting platform behind a large fleet of managed customer environments, applying AI-driven engineering to fleet operations at scale.
- Own platform architecture spanning Windows server fleets, Azure DevOps pipelines, and multi-tenant upgrade orchestration, plus relational schema design and maintenance.
- Built Sentry, unified monitoring for hosted databases and web servers: agentless collectors every 2 to 10 minutes, 100M+ samples in its first 34 days, and database health streamed to Azure with zero rejected points.
- Led GhostWatch, an AI-assisted incident pipeline started at the Microsoft Global Hackathon 2026: a Rust platform (1,200+ tests) that groups 18 live signal feeds into incidents. Rules proved 4x more accurate than a small LLM at labeling them, so rules lead and the LLM handles the tail.
- Maintain the environment inventory as the authoritative source of truth: 99% of its values collected automatically, and every other tool on the platform reads it before acting.
- Lead capacity and sizing analysis for executive decisions, and root-cause investigation on production incidents spanning application servers, storage, and reporting services.

03/2025 to 07/2026
Berwyn, PA

Operations Team Lead AssetWorks

- Built an internal AI agent platform on the Model Context Protocol: specialized LLM agents behind a custom semantic router (fine-tuned sentence-transformer embeddings, pgvector retrieval). 40,000+ routed requests since 03/2026, 83% handled without the LLM reasoning fallback.
- Agents write what they learn back to a shared store of 2,000+ knowledge patterns, so the platform improves with use.
- Built the Hosted Environment Navigator, the system of record for the hosted environment: 99% of its values collected automatically, behind a 500+ route API with 800+ tests.
- Built Hosted Access Manager for just-in-time, auto-expiring database access: 1,100+ time-boxed grants, each justified and audited, and 59% handed back before the timer ran out.
- Led a team of five; rolled out OpenTelemetry and Observe; mentored through 1:1s and training.

12/2022 to 01/2025
Boulder, CO

Backend Developer, Founder
EdventureTrek

- Founded an educational exploration game; designed custom taxonomy GPTs for plant and animal classification.
- Built the Python/FastAPI backend with MySQL and event logging; ran CI/CD on GCP with GitHub Actions.

07/2022 to 12/2022
Atlanta, GA

Site Reliability Engineering Manager
AnswerRocket

- Led a remote SRE team of four; ran weekly syncs and architecture reviews.
- Expanded Ansible coverage across AWS (15% less manual deploy time) and automated SOC 2 environment validation.

03/2016 to 07/2022
Alpharetta, GA

Site Reliability Architect
OfficeSpace Software

- Owned production and staging on GCP: patching, config management, release packaging, and automation with Puppet and Python.
- Migrated from Rackspace to GCP, saving \$60K annually; cut deploy times over 60% with a CircleCI, Puppet, Docker and Terraform pipeline.
- Built a Slackbot that deploys customer instances in under 10 minutes; hired, onboarded and led a three-person SRE team.

2009 to 2016
São Paulo, Brazil

Sr. Technical Consultant / Team Lead
Hewlett Packard

- Tier 3 support for HP Server Automation; automated workflows with Python against the HPSA API.
- Mentored junior engineers; ranked #1 in the team for customer satisfaction.



Education

10/2001 to 12/2004
Ft. Lauderdale, Florida

Information Systems | Bachelor's Degree
ITT Technical Institute
(Honors Graduate)



Strengths

Leadership AI-Led Ops Cloud Infrastructure Automation



Key Skills

CLOUD & AUTOMATION

GCP Azure Cloud Cloud Run Terraform Ansible Docker

CI/CD & DELIVERY

GitHub Actions Azure DevOps Jenkins

OBSERVABILITY

OpenTelemetry Prometheus PagerDuty Observe

AI & AGENTS

LLM Agents Model Context Protocol pgvector Hugging Face Claude

DEVELOPMENT & DATA

Python Redis Postgres

SYSTEMS & SERVING

Linux Nginx



Hobbies

Exploring AI tooling (LLMs, MCP) SRE meetups Home lab experimentation with Docker



Volunteering

07/2020 to 11/2024
Boulder, Colorado, USA

Delivery Driver / Wellness Check
Meals on Wheels Boulder

As a Meals on Wheels delivery driver, I got to enjoy great conversations with some of Boulder's greatest citizens.
mowboulder.org